

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
```

```
In [2]: df=pd.read_csv('D:\BMW sales data (2010-2024).csv')
```

```
In [3]: df.head()
```

```
Out[3]:
```

	Model	Year	Region	Color	Fuel_Type	Transmission	Engine_Size_L	Mileage_KM
0	5 Series	2016	Asia	Red	Petrol	Manual	3.5	151748
1	i8	2013	North America	Red	Hybrid	Automatic	1.6	121671
2	5 Series	2022	North America	Blue	Petrol	Automatic	4.5	10991
3	X3	2024	Middle East	Blue	Petrol	Automatic	1.7	27255
4	7 Series	2020	South America	Black	Diesel	Manual	2.1	122131



```
In [90]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Model                  50000 non-null object  
1   Year                   50000 non-null int64   
2   Region                 50000 non-null object  
3   Color                  50000 non-null object  
4   Fuel_Type              50000 non-null object  
5   Transmission           50000 non-null object  
6   Engine_Size_L          50000 non-null float64  
7   Mileage_KM             50000 non-null int64   
8   Price_USD              50000 non-null int64   
9   Sales_Volume           50000 non-null int64   
10  Sales_Classification   50000 non-null object  
dtypes: float64(1), int64(4), object(6)
memory usage: 4.2+ MB
```

```
In [5]: df.describe(include='all')
```

Out[5]:

	Model	Year	Region	Color	Fuel_Type	Transmission	Engine_Size_L
count	50000	50000.000000	50000	50000	50000	50000	50000.000000
unique	11	NaN	6	6	4	2	NaN
top	7 Series	NaN	Asia	Red	Hybrid	Manual	NaN
freq	4666	NaN	8454	8463	12716	25154	NaN
mean	NaN	2017.015700	NaN	NaN	NaN	NaN	3.247180
std	NaN	4.324459	NaN	NaN	NaN	NaN	1.009078
min	NaN	2010.000000	NaN	NaN	NaN	NaN	1.500000
25%	NaN	2013.000000	NaN	NaN	NaN	NaN	2.400000
50%	NaN	2017.000000	NaN	NaN	NaN	NaN	3.200000
75%	NaN	2021.000000	NaN	NaN	NaN	NaN	4.100000
max	NaN	2024.000000	NaN	NaN	NaN	NaN	5.000000

In [6]: `pd.isnull(df).sum()`

```
Out[6]: Model          0
Year          0
Region        0
Color         0
Fuel_Type     0
Transmission  0
Engine_Size_L  0
Mileage_KM     0
Price_USD     0
Sales_Volume  0
Sales_Classification  0
dtype: int64
```

In [93]: `Top_Models=df.groupby('Model')['Sales_Volume'].sum().sort_values(ascending=False)`In [94]: `Top_Models`

```
Out[94]: Model
7 Series    23786466
i8          23423891
X1          23406060
3 Series    23281303
i3          23133849
5 Series    23097519
M5          22779688
X3          22745529
X5          22709749
X6          22661986
M3          22349694
Name: Sales_Volume, dtype: int64
```

In [95]: `Sales_by_Region=df.groupby('Region')['Sales_Volume'].sum().sort_values(ascending`

```
In [96]: Sales_by_Region
```

```
Out[96]: Region
Asia          42974277
Europe        42555138
North America 42402629
Middle East   42326620
Africa         41565252
South America 41551818
Name: Sales_Volume, dtype: int64
```

```
In [7]: Yearly_Sales=df.groupby('Year')['Sales_Volume'].sum()
```

```
In [8]: Yearly_Sales
```

```
Out[8]: Year
2010    16933445
2011    16758941
2012    16751895
2013    16866733
2014    16958960
2015    17010207
2016    16957550
2017    16620811
2018    16412273
2019    17191956
2020    16310843
2021    16884666
2022    17920946
2023    16268654
2024    17527854
Name: Sales_Volume, dtype: int64
```

```
In [9]: df['Revenue']=df['Price_USD']*df['Sales_Volume']
```

```
In [10]: df['Revenue'].head(10)
```

```
Out[10]: 0      819542000
1      271562732
2      792175410
3      246749637
4      153685840
5       52884832
6      437703736
7      64955696
8     1041815008
9      424589627
Name: Revenue, dtype: int64
```

```
In [11]: Total_Revenue_by_Model=df.groupby('Model')['Revenue'].sum().sort_values(ascending=True)
```

```
In [12]: Total_Revenue_by_Model
```

```
Out[12]: Model
7 Series      1790070249282
3 Series      1768534028214
i8            1764743448529
X1            1752985285361
5 Series      1735712423092
i3            1724197530210
X5            1708653383772
X3            1707951188482
M5            1698467355916
X6            1693423254979
M3            1667504386622
Name: Revenue, dtype: int64
```

```
In [13]: Sales_by_Fuel_type=df.groupby('Fuel_Type')['Sales_Volume'].sum().sort_values(asc
```

```
In [14]: Sales_by_Fuel_type
```

```
Out[14]: Fuel_Type
Hybrid      64532097
Petrol      63324154
Electric    63157665
Diesel      62361818
Name: Sales_Volume, dtype: int64
```

```
In [15]: Sales_by_Transmission=df.groupby('Transmission')['Sales_Volume'].sum().sort_valu
```

```
In [16]: Sales_by_Transmission
```

```
Out[16]: Transmission
Manual      127362077
Automatic   126013657
Name: Sales_Volume, dtype: int64
```

```
In [17]: Avg_Price_by_Region=df.groupby('Region')['Price_USD'].mean().sort_values(ascendi
```

```
In [19]: Avg_Price_by_Region
```

```
Out[19]: Region
Asia      75554.925006
North America  75070.054709
Europe    74988.356851
South America  74973.598837
Africa    74885.771598
Middle East  74726.788487
Name: Price_USD, dtype: float64
```

```
In [20]: Premium_Sales=df[df['Price_USD']>df['Price_USD'].mean()]
```

```
In [21]: Premium_Sales
```

Out[21]:

	Model	Year	Region	Color	Fuel_Type	Transmission	Engine_Size_L	Mileage_k
0	5 Series	2016	Asia	Red	Petrol	Manual	3.5	1517
1	i8	2013	North America	Red	Hybrid	Automatic	1.6	1216
2	5 Series	2022	North America	Blue	Petrol	Automatic	4.5	109
7	M5	2014	Asia	Black	Diesel	Automatic	1.6	1211
8	X3	2016	South America	White	Diesel	Automatic	1.7	480
...	...	...	...	...	...	...	...	...
49982	M3	2024	Asia	Silver	Hybrid	Automatic	1.7	1988
49989	5 Series	2015	Africa	Grey	Electric	Manual	3.4	1480
49992	i3	2017	Middle East	Blue	Petrol	Automatic	3.9	798
49993	5 Series	2013	Europe	Grey	Petrol	Manual	4.7	1550
49999	X1	2020	North America	Blue	Diesel	Manual	3.3	1710

24989 rows × 12 columns



```
In [22]: Premium_Sales_by_Region=Premium_Sales.groupby('Region')['Sales_Volume'].sum().sort_values(ascending=False)
```

```
In [23]: Premium_Sales_by_Region
```

```
Out[23]: Region
Asia      22196954
North America  21303437
Europe    21090305
Middle East  20944114
South America  20683276
Africa     20678982
Name: Sales_Volume, dtype: int64
```

```
In [24]: High_sales=df[df['Sales_Classification']=='High']
```

```
In [25]: High_sales
```

Out[25]:

	Model	Year	Region	Color	Fuel_Type	Transmission	Engine_Size_L	Mileage_k
0	5 Series	2016	Asia	Red	Petrol	Manual	3.5	1517
6	i8	2022	Europe	White	Diesel	Manual	1.8	1967
8	X3	2016	South America	White	Diesel	Automatic	1.7	480
11	i8	2016	North America	Blue	Hybrid	Manual	1.9	1168
12	7 Series	2020	North America	Silver	Diesel	Automatic	3.8	274
...	...	...	...	...	...	...	...	...
49992	i3	2017	Middle East	Blue	Petrol	Automatic	3.9	798
49995	i3	2014	Asia	Red	Hybrid	Manual	4.6	1510
49996	i3	2023	Middle East	Silver	Electric	Manual	4.2	1473
49997	5 Series	2010	Middle East	Red	Petrol	Automatic	4.5	1749
49998	i3	2020	Asia	White	Electric	Automatic	3.8	33

15246 rows × 12 columns

In [26]: `Low_sales=df[df['Sales_Classification']=='Low']`In [27]: `Low_sales`

Out[27]:

	Model	Year	Region	Color	Fuel_Type	Transmission	Engine_Size_L	Mileage_k
1	i8	2013	North America	Red	Hybrid	Automatic	1.6	1216
2	5 Series	2022	North America	Blue	Petrol	Automatic	4.5	109
3	X3	2024	Middle East	Blue	Petrol	Automatic	1.7	272
4	7 Series	2020	South America	Black	Diesel	Manual	2.1	1221
5	5 Series	2017	Middle East	Silver	Diesel	Manual	1.9	1713
...	...	...	...	...	...	...	...	...
49988	X5	2013	South America	Grey	Hybrid	Manual	3.9	1775
49991	X5	2019	Asia	White	Diesel	Automatic	3.3	1626
49993	5 Series	2013	Europe	Grey	Petrol	Manual	4.7	1550
49994	5 Series	2010	North America	Silver	Petrol	Manual	3.6	566
49999	X1	2020	North America	Blue	Diesel	Manual	3.3	1710

34754 rows × 12 columns



In [28]: `Sales_by_Engine_Size=df.groupby('Engine_Size_L')['Sales_Volume'].sum().sort_valu`

In [29]: `Sales_by_Engine_Size.head(10)`

Out[29]:

Engine_Size_L	
1.9	7667354
4.5	7657060
3.8	7573254
2.5	7560056
3.2	7495239
2.7	7488867
4.2	7482993
4.7	7432100
2.6	7423588
3.4	7416505

Name: Sales_Volume, dtype: int64

In [30]: `Price_and_Sales_Correlation=df['Price_USD'].corr(df['Sales_Volume'])`

In [31]: `Price_and_Sales_Correlation`

Out[31]: `np.float64(7.95632646798523e-05)`

```
In [32]: Sales_by_Colour=df.groupby('Color')['Sales_Volume'].sum().sort_values(ascending=
```

```
In [33]: Sales_by_Colour
```

```
Out[33]: Color
Red      42750183
Silver   42674022
White    42272954
Grey     41995141
Blue     41972741
Black    41710693
Name: Sales_Volume, dtype: int64
```

```
In [34]: from sklearn.linear_model import LinearRegression

X=df[['Year']]
y=df['Sales_Volume']

model=LinearRegression()
model.fit(X,y)

future_years=pd.DataFrame({'Year': [2025,2026,2027,2028]})
predictions = model.predict(future_years)
```

```
In [35]: print(predictions)
```

```
[5076.32701875  5077.43072712  5078.53443549  5079.63814386]
```

```
In [36]: df['Sales_Category']=pd.cut(
    df['Sales_Volume'],
    bins=[0,500,2000,5000,10000],
    labels=['Low', 'Average', 'Medium', 'High']
)
```

```
In [37]: df[['Sales_Volume', 'Sales_Category']].head(10)
```

```
Out[37]:
```

	Sales_Volume	Sales_Category
0	8300	High
1	3428	Medium
2	6994	High
3	4047	Medium
4	3080	Medium
5	1232	Average
6	7949	High
7	632	Average
8	8944	High
9	4411	Medium

```
In [38]: df['Price_Category']=pd.cut(
df['Price_USD'],
bins=[0,40000,80000,150000],
labels=['Standard', 'Premium', 'Elite']
)
```

```
In [39]: df[['Price_USD', 'Price_Category']].head(10)
```

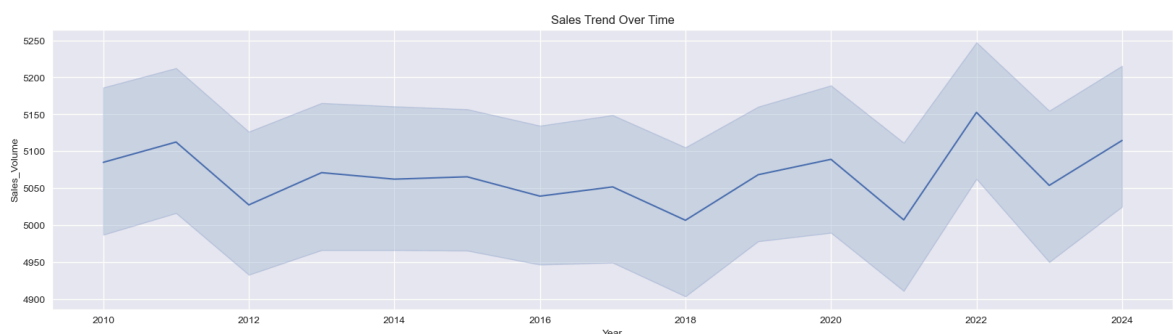
```
Out[39]:
```

	Price_USD	Price_Category
0	98740	Elite
1	79219	Premium
2	113265	Elite
3	60971	Premium
4	49898	Premium
5	42926	Premium
6	55064	Premium
7	102778	Elite
8	116482	Elite
9	96257	Elite

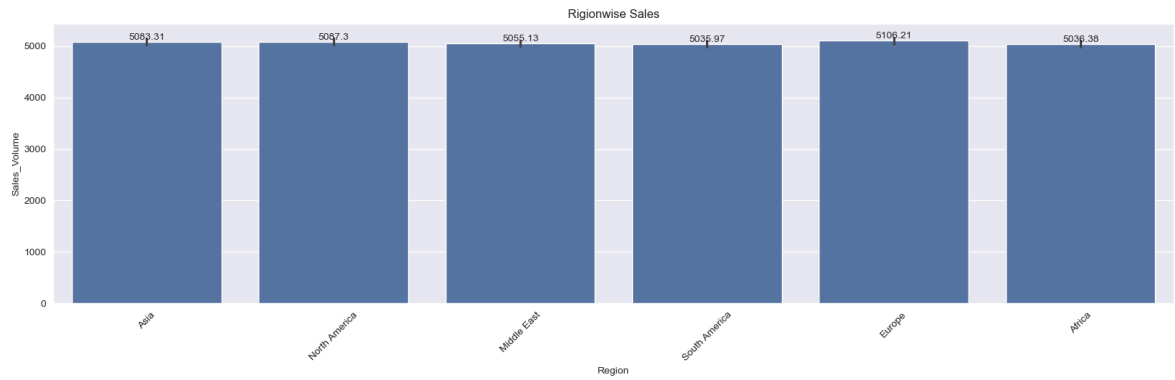
```
In [40]: df.columns
```

```
Out[40]: Index(['Model', 'Year', 'Region', 'Color', 'Fuel_Type', 'Transmission',
'Engine_Size_L', 'Mileage_KM', 'Price_USD', 'Sales_Volume',
'Sales_Classification', 'Revenue', 'Sales_Category', 'Price_Category'],
dtype='object')
```

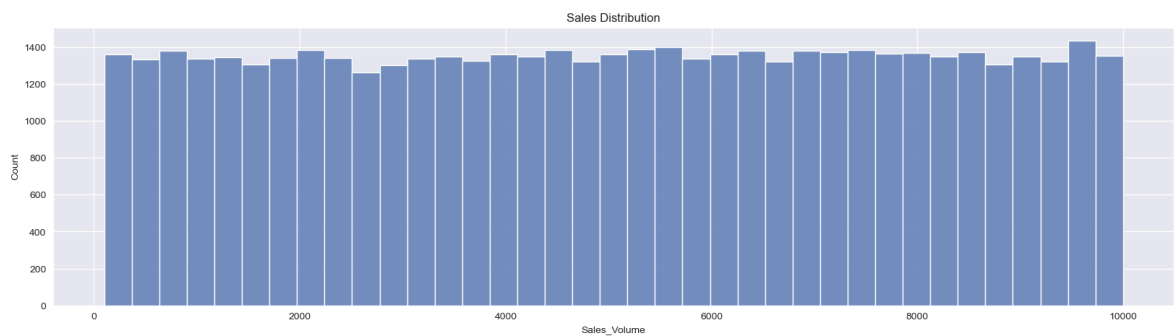
```
In [53]: sns.lineplot(x='Year', y='Sales_Volume', data=df)
plt.title('Sales Trend Over Time')
sns.set({'figure.figsize':(20,5)})
plt.show()
```



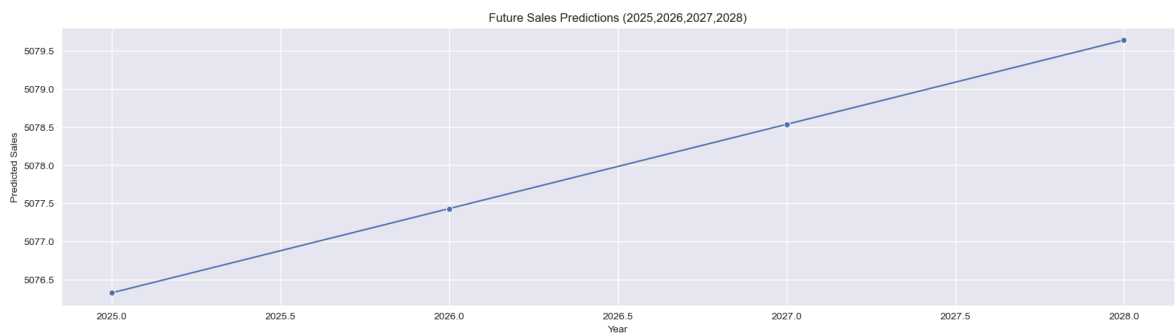
```
In [55]: ax=sns.barplot(x='Region',y='Sales_Volume',data=df)
for bars in ax.containers:ax.bar_label(bars)
sns.set({'figure.figsize':(15,5)})
plt.xticks(rotation=45)
plt.title('Rigionwise Sales')
plt.show()
```



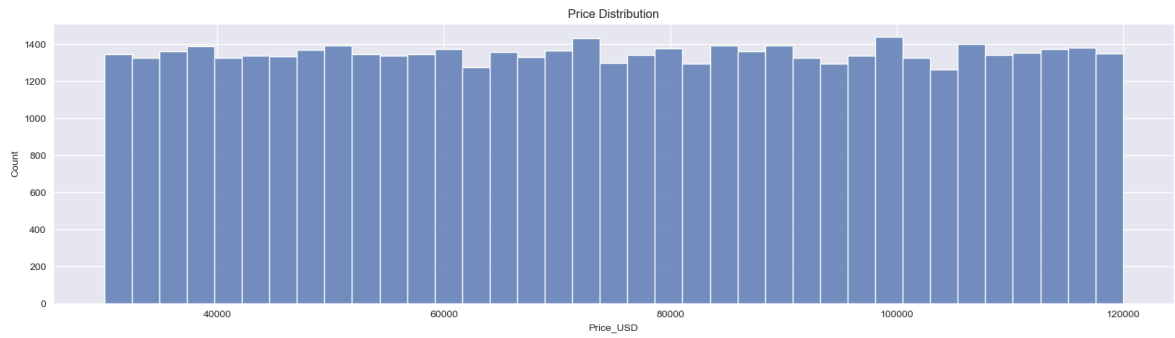
```
In [52]: sns.histplot(df['Sales_Volume'])
plt.title('Sales Distribution')
sns.set({'figure.figsize':(20,5)})
plt.show()
```



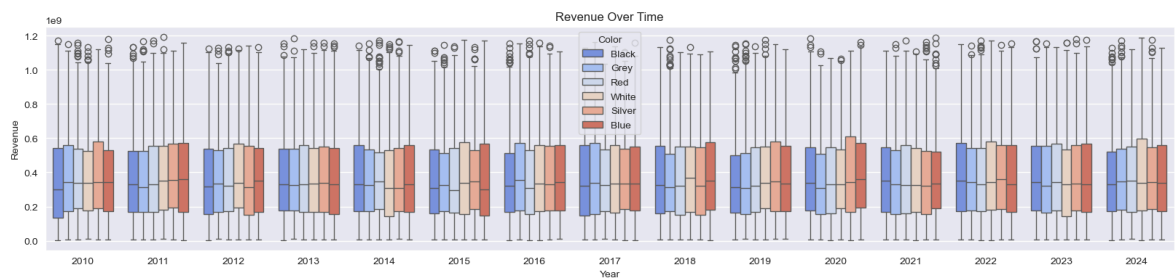
```
In [84]: sns.lineplot(x=future_years['Year'], y = predictions, data=future_years, marker=
plt.title('Future Sales Predictions (2025,2026,2027,2028)')
plt.xlabel('Year')
plt.ylabel('Predicted Sales')
sns.set({'figure.figsize':(20,5)})
plt.show()
```



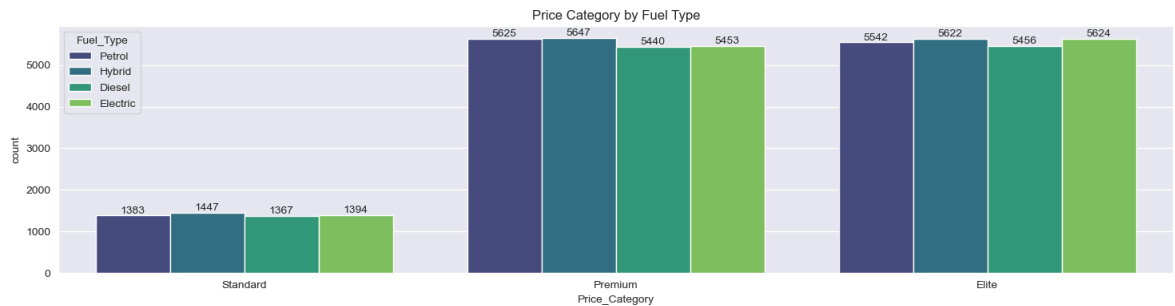
```
In [51]: sns.histplot(df['Price_USD'])
plt.title('Price Distribution')
sns.set({'figure.figsize':(20,5)})
plt.show()
```



```
In [78]: sns.boxplot(x='Year',y='Revenue',hue='Color',palette='coolwarm',data=df)
plt.title('Revenue Over Time')
sns.set({'figure.figsize':(18,4)})
plt.show()
```



```
In [79]: ax=sns.countplot(x='Price_Category',hue='Fuel_Type',palette='viridis',data=df)
for bars in ax.containers: ax.bar_label(bars)
plt.title('Price Category by Fuel Type')
sns.set({'figure.figsize':(20,4)})
plt.show()
```



```
In [92]: df.to_csv('BMW_Sales_Data_Cleaned_(2010-2024).csv',index=False)
```

```
In [ ]:
```